

PROJECT DATA SHEET



CASE Consultants

Civil And Structural Engineers

Moose Hall

Barewell Road

Torquay

TQ1 4PA

tel. (01803) 200175

fax. (01803) 323205

E-Mail: admin@caseconsultants.net

JOB/PROJECT : NOSS ON DART MARINA, DARTMOUTH

JOB/PROJECT NO : C8829 SHEET NO : COVER

GENERAL :

CLIENT : PREMIER MARINAS

ARCHITECT : N/A

AUTHORITY : TORBAY COUNCIL

JOB/PROJECT DESCRIPTION : APARTMENT BLOCK FALSEWORK SUPPORT TO UNDERCROFT

PROPOSED USE: PROP BASE SPREADERS

OVERALL SIZE : AS CALCS

No. OF STOREYS : N/A

TYPE OF SOIL INVESTIGATION : PLATE BEARING TEST

BEARING STRATA : COMPACTED 803 MATERIAL

TYPE OF FOUNDATION : C24 TIMBER SPEADERS

MAXIMUM SAFE BEARING PRESSURE 250 kN/m²

STRUCTURAL MATERIALS USED : DOKA EUREX 30 PROPS

STANDARDS AND CODES USED IN DESIGN :

General actions: Densities, self weight & imposed loads	BS EN 1991-1-1	☒	Steelwork	BS EN 1993 (EC3)	☐
General actions: Actions on structures exposed to fire	BS EN 1991-1-2	☐	Timber	BS EN 1995 (EC5)	☐
			Masonry	BS EN 1996 (EC6)	☐
Snow Loads	BS EN 1991-1-3	☐	Others	DOKA DATASHEET	☒
Wind Loads	BS EN 1991-1-4	☐			☐
Concrete	BS EN 1992 (EC2)	☐			☐

AUTHORISATION OF CALCULATION SHEETS AND BENDING SCHEDULES :

CALCULATION SHEET NOS.	CHECKED BY	DATE
Sheet nos: S01 - S06 inclusive	<i>Clough</i>	22/04/25.



Calculations

Contract	<u>NOSS ON DART MARINA</u> <u>DARTMOUTH</u>	Job No.	<u>C8829</u>	Sheet No.	<u>501</u>
		Designed by	<u>A.C.H.</u>	Date:	<u>APRIL 25</u>

APARTMENT SHORING

These calculations cover the design of the base spreaders to the soffit propping to the apartments between gridlines T & U. These props are to be supported onto a compacted granular sub-base. Plate bearing tests have been carried out on the sub-base to establish the load capacity.

Taking an ultimate deflection value of 5.0mm the lowest pressure recorded is 500 kN/m^2 .

Applying a factor of safety of 2.0 gives an allowable S.G.B.P of 250 kN/m^2 .

Determine prop loads.

For the general case the spacing of primary beams is 2.0m with props at 1.0m cs along their length.

Global load is as follows.

$$275 \text{ kN/m}^2 \text{ concrete} = 24 \times 0.275 = 6.60 \text{ kN/m}^2$$

$$\text{Construction load} = 2.00 \text{ kN/m}^2$$

$$\Sigma = 8.60 \text{ kN/m}^2 \text{ say } 9.00 \text{ kN/m}^2 \text{ incl formwork.}$$

$$\text{Characteristic prop load} = 9 \times 2 \times 1 \\ = 18 \text{ kN.}$$

$$\therefore \text{Area Req'd} = \frac{18}{250} = 0.072 \text{ m}^2$$



Calculations

Contract

NOSS ON DART MARINA
DARTMOUTH

Job No.

C8829

Sheet No. 502

Designed by

A. C. H.

Date: APRIL 25

with $b = 220 \text{ mm}$.

length reqd = $0.072 / 0.22 = 0.327 \text{ mm}$. $\therefore$ USE 450.

For subsequent floors it is taken that there will be back propping to the lower level upto casting level 3 slab ie 2 No. levels of back-propping below the slab supporting the soffit shutter. This is as set out in Gallagher method statement ref MS 791/18 page 6.

The maximum prop load will therefore be as calculated when the props are supporting the shutter directly.

$\therefore$, FOR GENERAL PROPS TO 1ST FLOOR
SOFFIT SHUTTER PROVIDE 220x72x450g C14
TIMBER SPREADER ONTO WELL COMPACTED
803 SUB-BASE BETWEEN GRIDLINE T&U.



Calculations

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		Designed by	<u>A.C.H.</u>	Date:	<u>APRIL 25</u>

BALCONY PROPS

The corner balcony between grids 8 & 9 is supported with a thermal break system that cannot support the construction loads. ∴ The back propping will be designed to carry the full load of the 1st, 2nd & 3rd floor slabs.

LOADING.

$$\text{Concrete weight} = 3 \times 0.275 \times 24 = 19.8 \text{ kN/m}^2$$

$$\text{Construction Load} = 2.0 \text{ kN/m}^2$$

$$\text{Access load on lower levels} = 2.0 \times 0.5 = 1.0 \text{ kN/m}^2$$

$$\underline{\underline{\Sigma = 22.8 \text{ kN/m}^2}}$$

with 3.0m max strut length

SWL for Doka 30 = 30kN

$$\therefore \text{Maximum area} = 30 / 22.8 = 1.3 \text{ m}^2 = 1.14 \text{ m cs.}$$

∴ Try 1.0m cs max.

Maximum prop load = 22.8kN

$$\text{Contact area onto stone} = 28 / 250 = 0.112 \text{ m}^2$$

$$\text{with } b = 220 \text{ mm } l_{\text{reqd}} = 0.112 / 0.22 = 0.51 \text{ m}$$

Try 600mm long.



Calculations

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		Designed by	A.CH.	Date:	APRIL 25

$$\text{Max bearing pressure} = \frac{22.8}{0.22 \times 0.6} = 172 \text{ kN/m}^2 \text{ char}$$

$$= \underline{\underline{242 \text{ kN/m}^2 \text{ ok}}}$$

$$M_d = (242 \times 0.22) \times 0.3^2 / 2 = 2.40 \text{ kN/m}^2$$

$$\sigma_b \text{ perm} = 24 \times 0.65 = 15.6 \text{ N/mm}^2$$

k_{mod} - class 3 medium term

$$Z_{yy} \text{ Regd} = \frac{2.40 \times 10^6}{15.6} = 153 \times 10^3 \text{ mm}^3$$

with $b = 220 \text{ mm}$

$$d_{regd} = \sqrt{\frac{b \times 153 \times 10^3}{220}} = 64 \text{ mm.} \therefore 75 \text{ mm ok}$$

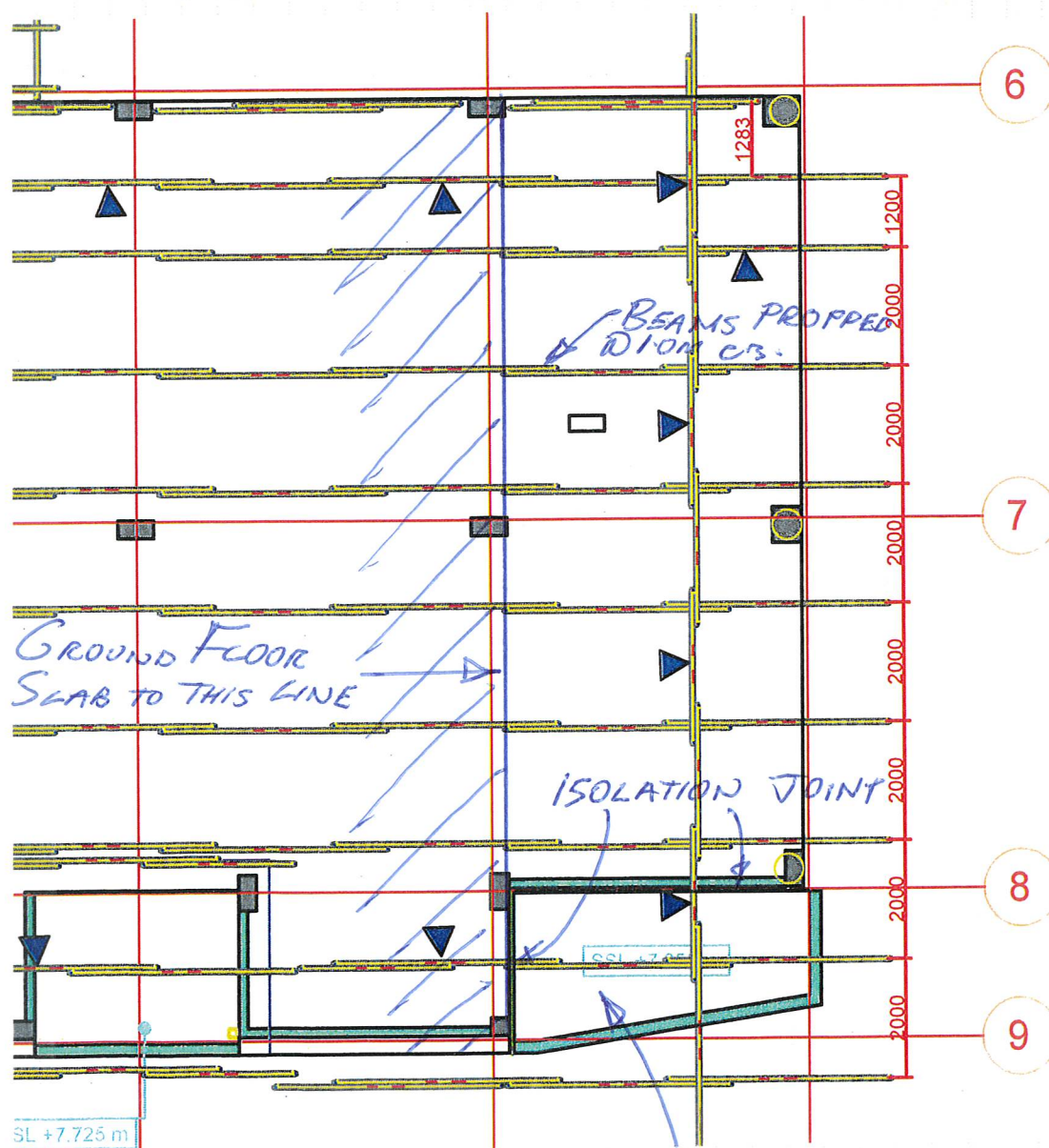
∴ Use Doka Euxex 30 PROPS
@ 1.0m CS AT GROUND FLOOR LEVEL,
AS BACK PROPS ONTO 220x72x600ly
C24 TIMBER SPREADER ONTO WELL
COMPACTED 803 SUB-BASE TO SUPPORT
CANTILEVERED BALCONY BETWEEN GRIDS
8-9.



Calculations

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		Designed by	<u>A.C.H.</u>	Date:	<u>APRIL 25</u>

EXTRACT OF GALLAGHER DRG No. DAR-MTG-DR-XX163



GENERALLY USE
72x220 x 450 Lg
C 24 SPREADER
ONTO SUB-BASE,
BELOW EUREX 30
PROPS.

CORNER BALCONY TO
BE BACK PROPPED @
1.0M CS. MAX
(USE 72x220 x 600 Lg
C 24 SPREADERS ONTO
SUB-BASE).



Calculations

Contract

Noss on Dart
DARTMOUTH

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Sheet No.

506

Designed by

A.C.H.

Date:

APRIL 25

DOKA EUREX 30 CAPACITY TABLE

Position of outer tube		Eurex 20 top										Eurex 30 top												
		150	250	300	350	400	550	250	300	350	400	450	550											
at bot- tom	at top	Top or bottom	at bottom	at top	at bottom	at top	at bottom	at top	at bottom	at top	at bottom	at top	at bottom	at top										
		Prop category to EN 1065	D15	B25 D25	B30 D30	C30 D30	C35 D35	C40 D40	C40 D40	C55 D55	C55 D55	C25 E25	C25 E25	C30 E30	C30 E30	C35 E35	C35 E35	C40 E40	C40 E40	C45 E45	C45 E45	C55 E55	C55 E55	
	Prop length [m]	5.5								20.6	22.7											31.8	33.3	
		5.4									21.6	23.9										33.6	35.3	
		5.3									22.5	25.2										35.5	37.2	
		5.2									23.6	26.5										37.3	39.2	
		5.1									24.7	27.9										39.2	41.0	
		5.0									25.8	29.4												
		4.9									27.2	31.0												
		4.8									28.7	32.5												
		4.7									30.1	34.2												
		4.6									31.6	35.9												
		4.5									33.2									32.7	34.5			
		4.4									34.9									34.8	36.8			
		4.3																		36.8	39.2			
		4.2																		39.2				
		4.1																						
		4.0							21.5	24.8							31.5	34.2					41.2	41.2
		3.9							23.0	26.8							33.8	36.8						
		3.8							24.4	28.7							36.1	39.3						
		3.7							26.0	30.8							38.7							
		3.6							27.7	33.2														
		3.5					20.8	24.5	29.4	35.5							30.9	34.2						
		3.4					22.3	26.7	31.0								33.3	36.8						
		3.3					23.8	28.9	32.4								35.8	39.3						
		3.2					25.4	31.3	33.5								38.2							
		3.1					27.1	34.0	34.5															
		3.0				20.7	24.8	28.8	35.5															
		2.9				22.4	27.4	31.5	36.7															

Noss on Dart - Balcony Propping Consideration (back propping)

G.Tonks - 791-SK-B-01 - 21-01-25 - Rev 00

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Level 01

Perimeter Prop load 23 kN

Max Prop load (internal) 12 kN

No structural slab at ground floor level,
area needs to be prepared for balcony
prop loads

